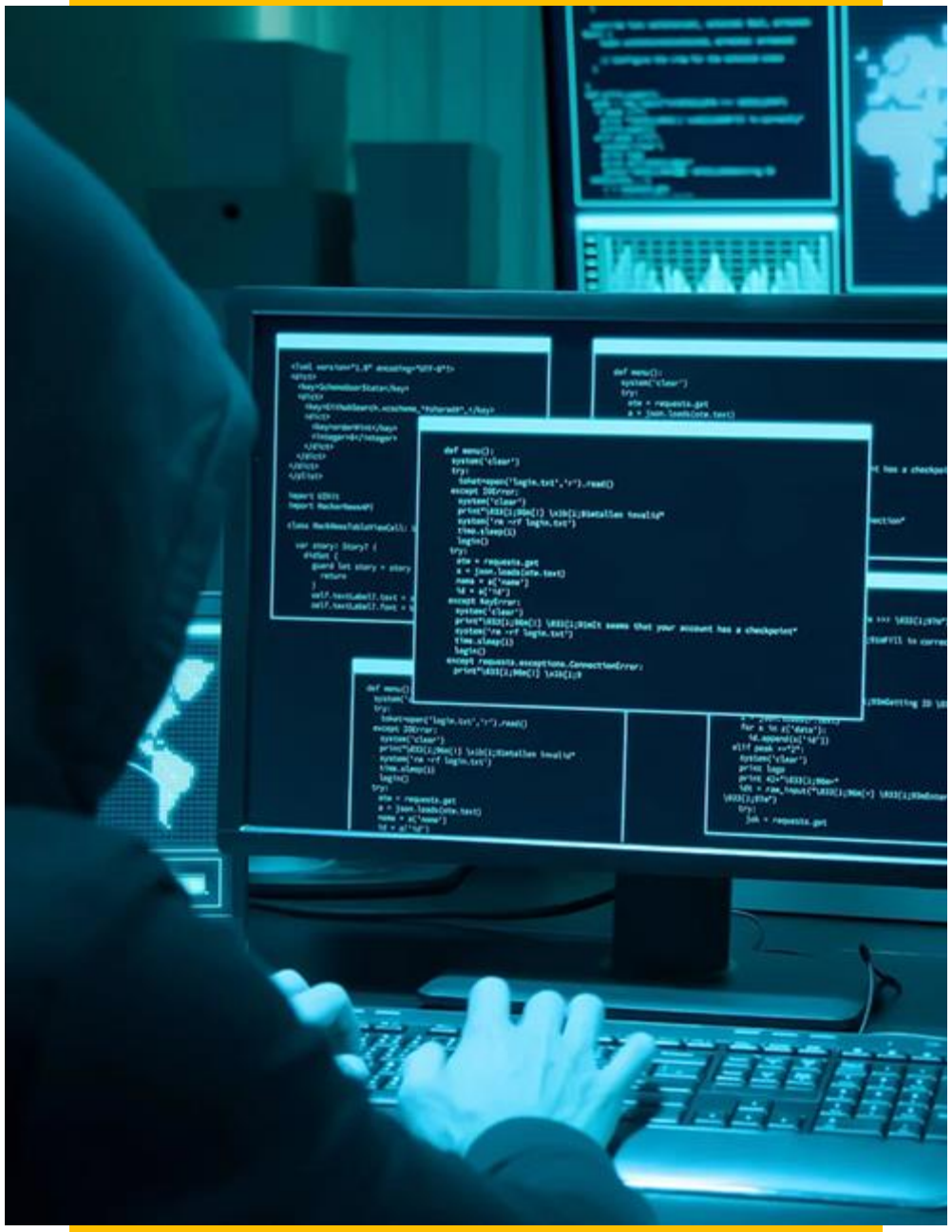


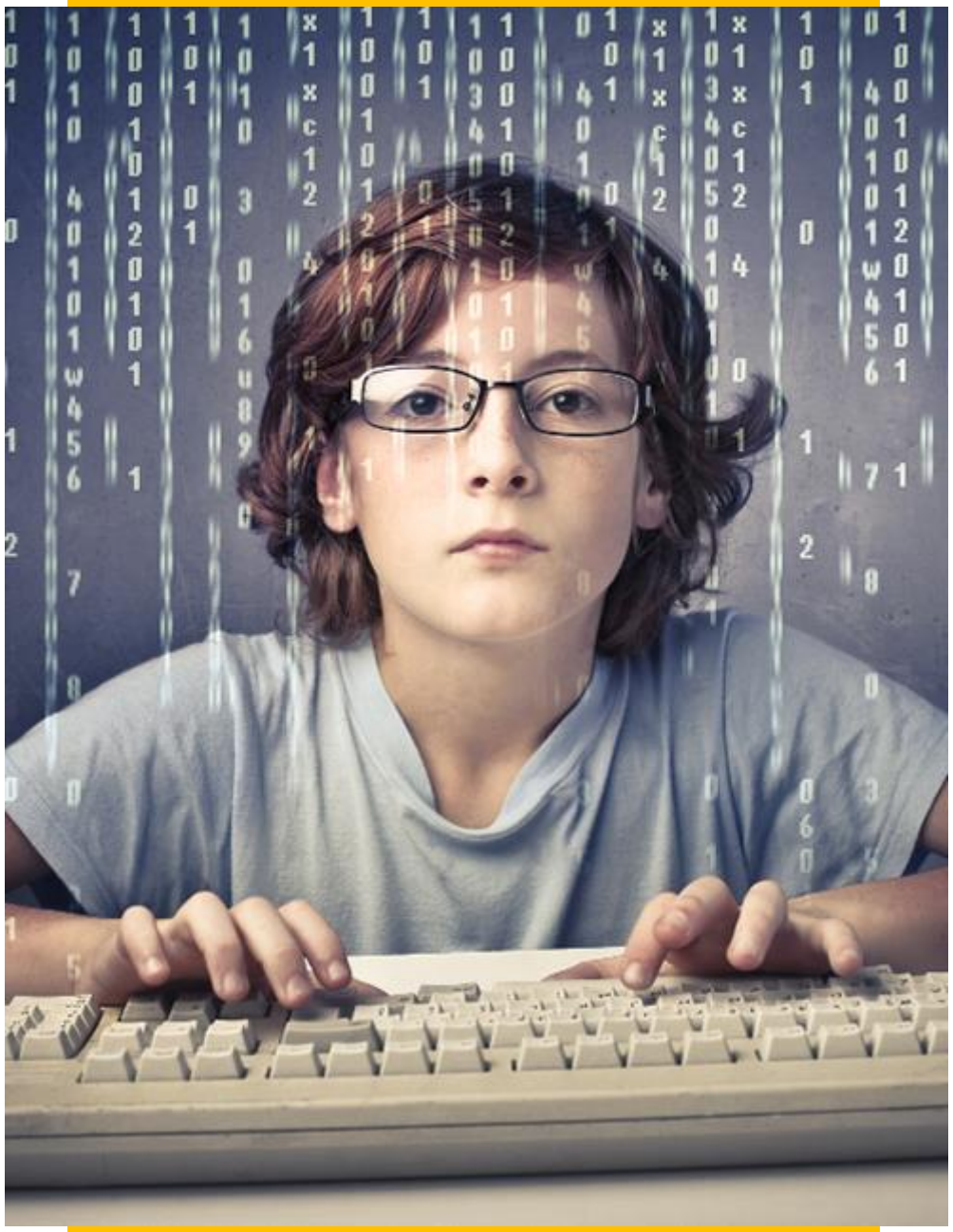
CYBERSECURITY THREATS

- ❑ Attacker profiles and motivations
- ❑ Modern cyber-attack strategy
- ❑ Wi-Fi Configuration and Attacks
- ❑ Advanced Persistent Threats



ATTACKER PROFILES AND MOTIVATIONS

- Script Kiddies
- Hacktivists
- Criminals
- State-Sponsored Hackers
- Insiders
- Organized Cybercrime Groups
- Cyberterrorists



SCRIPT KIDDIES

Characteristics of script kiddies:

- ❑ Limited Technical Knowledge
- ❑ Use of Automated Tools
- ❑ Minimal Originality
- ❑ Target Easier Targets
- ❑ Limited Scope
- ❑ Lack of Ethics



HACKTIVISTS

- ❑ Motivated by Ideals
- ❑ Political or Social Activism
- ❑ Civil Disobedience
- ❑ Anonymity
- ❑ Publicity and Attention
- ❑ Ethical Considerations
- ❑ Varied Techniques
- ❑ Diverse Groups



CYBER CRIMINALS

- ❑ Hacking
- ❑ Malware Distribution
- ❑ Phishing
- ❑ Identity Theft
- ❑ Online Fraud
- ❑ Distributed Denial of Service (DDoS) Attacks
- ❑ Data Breaches
- ❑ Cyber Extortion and Espionage
- ❑ Black Market Activities



STATE-SPONSORED HACKERS

- ❑ Sophistication
- ❑ Target Diversity
- ❑ Espionage
- ❑ Geopolitical Motivations
- ❑ Attribution Challenges
- ❑ Advanced Persistent Threats
- ❑ Criminal and Strategic Objectives
- ❑ Implications



INSIDERS

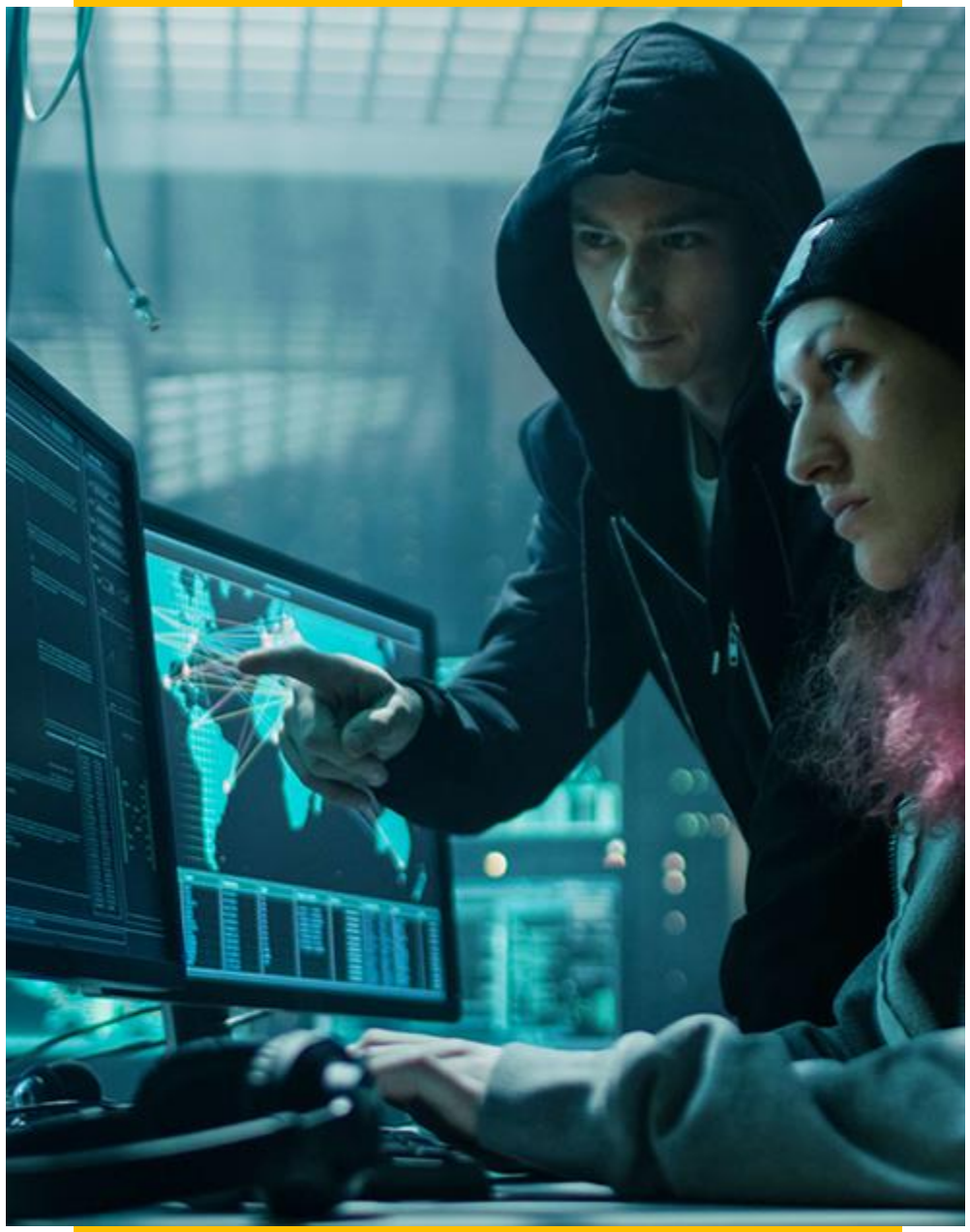
- Individuals who have authorized access to an organization's systems, networks, data, or facilities due to their status as employees
- These individuals, by virtue of their access, have the potential to pose security risks if they misuse their privileges, intentionally or unintentionally



INSIDERS

Key Strategies to avoid Insiders Attack

- ❑ Access Controls
- ❑ Security Awareness Training
- ❑ Monitoring and Auditing
- ❑ Incident Response Planning
- ❑ Employee Support



ORGANIZED

CYBERCRIME GROUPS

- Organized cybercrime groups are highly structured
- Doing for financial gain, power, or other objectives
- These groups use advanced hacking techniques
- They pose significant threats to individuals, organizations, and even governments
  - Russian Cybercriminal Syndicates
  - Chinese APT Groups
  - North Korean Groups



CYBERTERRORISTS

- Cyberterrorists are individuals or groups that use cyber-attacks
- Cyberterrorism involves using information technology to target critical infrastructure
- The goal of cyberterrorism is to create chaos, panic, and disruption within a society
  - Political or Ideological Motivation
  - Targeting Critical Infrastructure
  - Technological Sophistication



CYBER-ATTACK STRATEGY



UN AUTHORIZED ACCESS

- ❑ Reconnaissance
- ❑ Weaponization
- ❑ Exploitation



CYBER-ATTACK STRATEGY

- It is the act of utilizing computer systems, networks, data, or resources without proper authorization or permission
- This term is often associated with malicious activities carried out by attackers
  - Installation
  - Command and Control
  - Actions on the objectives



RECONNAISSANCE

- It refers to the preparatory phase of an attack, where an adversary gathers information about a target system, network, organization
- This initial stage is crucial for attackers to plan their strategies and tailor their attacks effectively
  - PASSIVE RECONNAISSANCE
  - ACTIVE RECONNAISSANCE



WEAPONIZATION

- It refers to the process of converting vulnerabilities into malicious payloads or cyber weapons
- The goal is to create and executing the attack on a target system
  - Malicious Payloads
  - Exploitation of Vulnerabilities
  - Delivery Mechanisms
  - Social Engineering
  - Customization



EXPLOITATION

- Cyber attacker takes advantage of a vulnerability in a system to gain unauthorized access,
- It execute malicious code or achieve specific objectives.
  - Vulnerability Exploitation
  - Unauthorized Access
  - Delivery of Payload
  - Privilege Escalation
  - Post-Exploitation Actions



INSTALLATION

- Installation refers to the phase of a cyber attack where an attacker establishes a foothold on a compromised system.
- The installation phase is a critical step for the attacker
  - Persistence
  - Backdoors
  - Remote Access
  - Privilege Escalation
  - Lateral Movement
  - Mitigation



COMMAND AND CONTROL

- It(C2) refers to a centralized infrastructure or mechanism used by cyber attackers.
- It serves as a communication channel between the attacker and the compromised systems
  - Botnets
  - Post-Exploitation Actions
  - Persistence
  - Detection and Mitigation



ACTIONS ON THE OBJECTIVES

It refers to the cyber attacker carries out once they have successfully achieved their primary goals or objectives within a targeted system.

- Data Exfiltration
- Lateral Movement
- Privilege Escalation
- Staging Attacks
- Damage and Disruption
- Covering Tracks
- Continued Persistence



WI-FI CONFIGURATION AND ATTACKS

- ❑ SSID (Service Set Identifier)
- ❑ Security Protocols
- ❑ Password
- ❑ Network Mode
- ❑ Channel Selection
- ❑ Guest Network
- ❑ MAC Address Filtering
- ❑ Remote Management



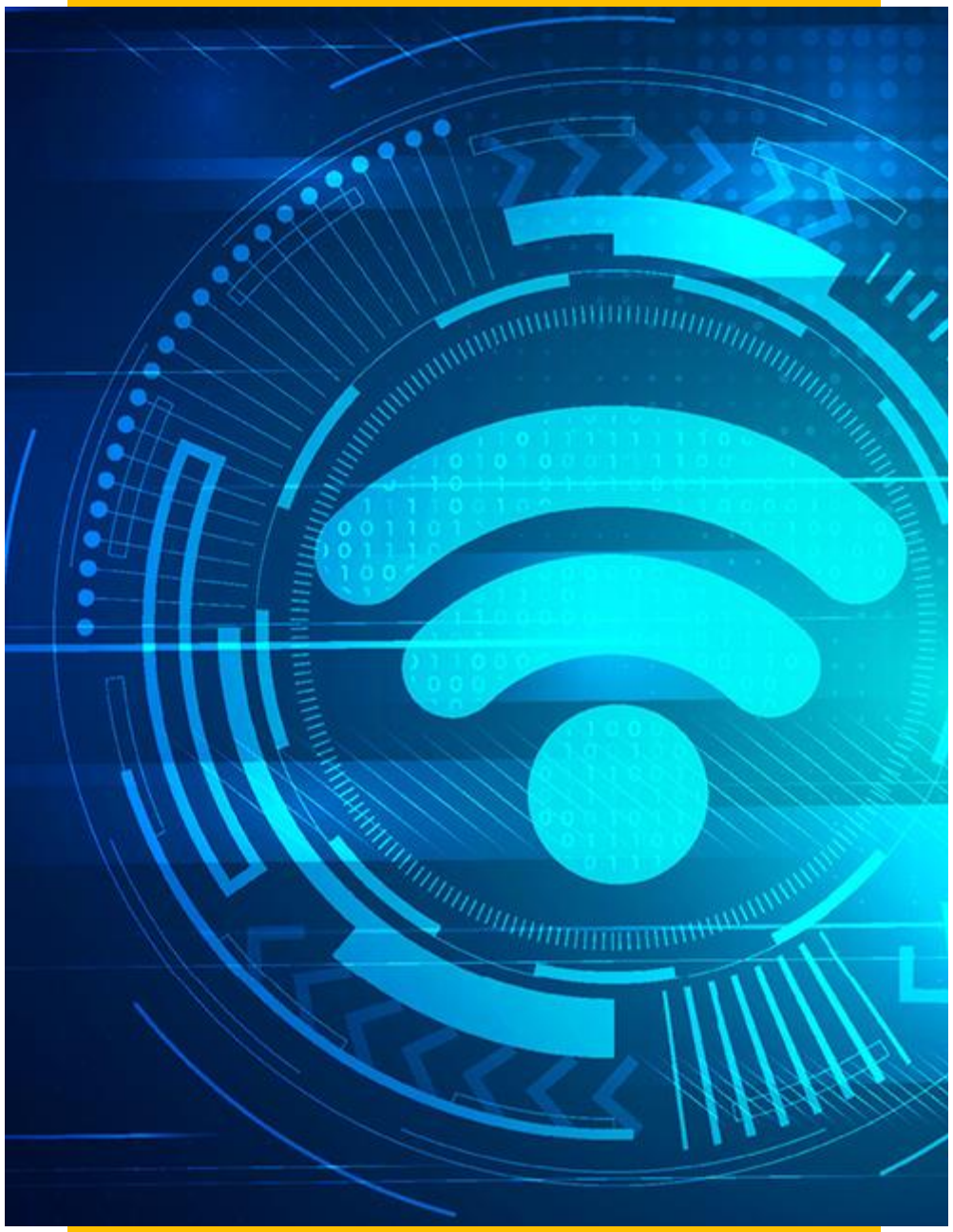
SSID (SERVICE SET IDENTIFIER)

- An SSID is a unique name assigned to a wireless network to identify it within a specific local area
- It helps users distinguish between different wireless networks in their vicinity
  - Broadcasting
  - Security Implications
  - Network Configuration
  - SSID Spoofing



SECURITY PROTOCOLS

- These are sets of rules and procedures to define how to establish secure communication
  - Encryption Protocols
  - Authentication Protocols
  - Key Exchange Protocols
  - Integrity Protocols
  - Access Control Protocols
  - Network Security Protocols
  - Secure Communication Protocols
  - Wireless Security Protocols



PASSWORD

They serve as a form of access control, helping to safeguard sensitive information and prevent unauthorized access.

- User Authentication
- Security
- Password Policy
- Multi-Factor Authentication (MFA)
- Password Management
- Account Recovery
- Password Attacks
- Password Hashing



NETWORK MODE

The network mode defines how the device functions in terms of routing, security, communication, and connectivity.

- Router Mode
- Bridge Mode
- Access Point (AP) Mode
- Client Mode
- Gateway Mode
- Monitor Mode
- Promiscuous Mode



WI-FI ATTACKS

That can arise from improper configuration or security weaknesses

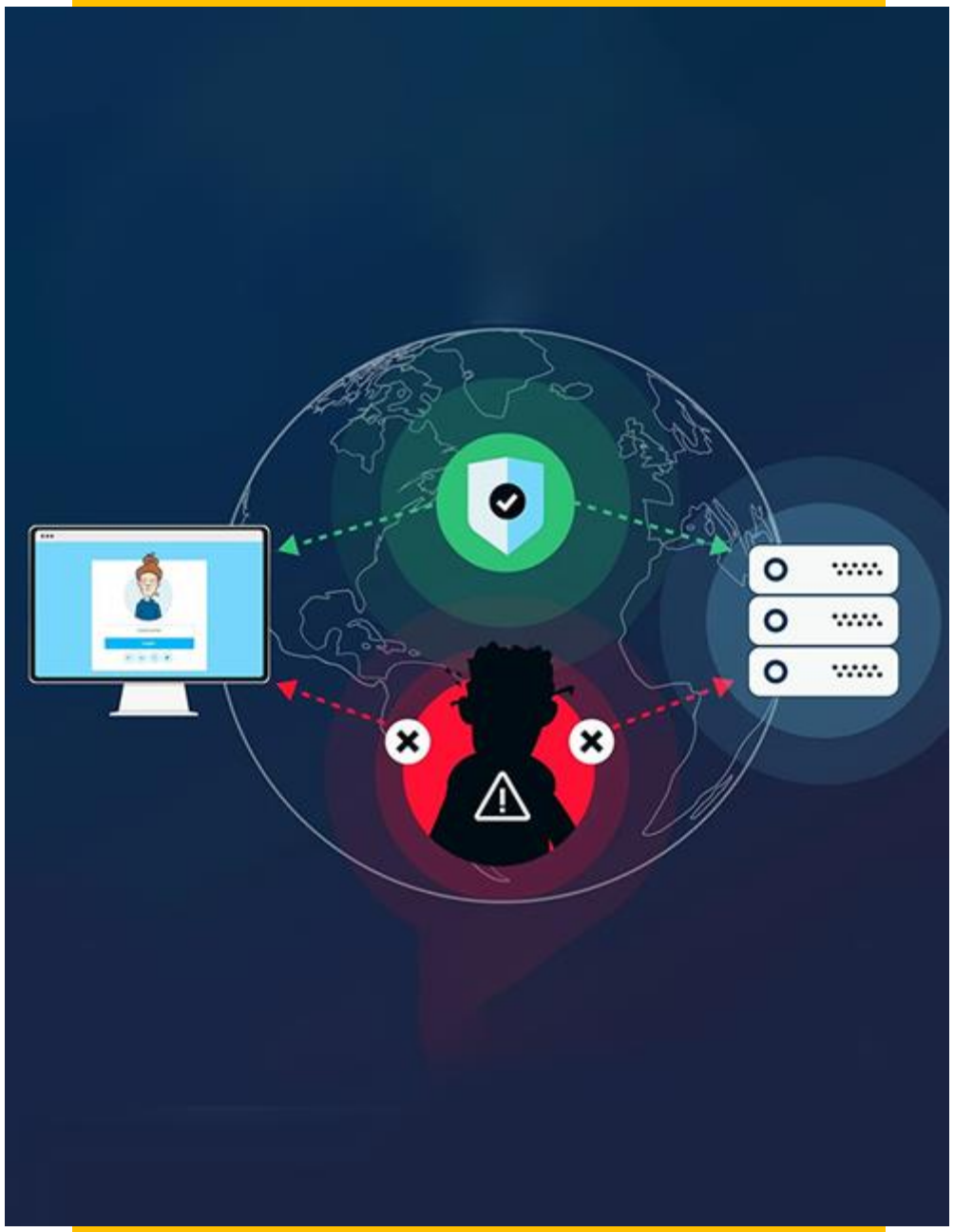
- Rogue Access Points
- Man-in-the-Middle (MitM)
- Evil Twin
- Password Cracking
- De authentication Attacks
- Eavesdropping
- WPS (Wi-Fi Protected Setup) Vulnerabilities
- KRACK Attack



ROGUE ACCESS POINTS

Rogue access point in cybersecurity refers to an unauthorized wireless access point (AP)

- Unauthorized Deployment
- Wireless Interference
- Security Risks
- Network Segmentation
- MitM Attacks
- Unauthorized Guests
- Detection
- Prevention and Security Measures



MAN-IN-THE-MIDDLE (MITM)

Attacker secretly intercepts, alters, or relays communication between two parties

- Attack Vector
- Interception
- Modification
- Credentials Theft
- Encryption Bypass
- Public Wi-Fi
- Protective Measures
- Digital Certificates



EVIL TWIN

The evil twin is set up to mimic a trusted network, but it is controlled by the attacker

- Deception
- Interception
- Man-in-the-Middle
- Credential Theft
- Phishing
- Protection
- Rogue Access Points
- Detection



PASSWORD CRACKING

The goal of password cracking is to overcome the security of a protected resource

- Brute Force
- Dictionary Attack
- Rainbow Tables
- Credential Stuffing
- Cracking Hashed Passwords
- Salting
- Rainbow Table Defenses
- Security Implications



DE AUTHENTICATION ATTACKS

It is a type of wireless network attack that involves sending forged de authentication frames to wireless devices within a network

- Wireless Disruption
- Unauthorized Disconnection
- Disconnecting Users
- Re-Authentication
- MitM Attack Amplification



EAVESDROPPING

Eavesdropping is capturing and listening to the communication of individuals or organizations without their knowledge or consent

- Passive Monitoring
- Wiretapping
- Data Interception
- Attack Methods
- Privacy Violation
- Secure Wi-Fi
- Monitoring for Defense



HOW TO AVOID WIFI ATTACKS

- Use strong security protocols like WPA3
- Regularly update router firmware to patch vulnerabilities
- Choose strong, unique Wi-Fi passwords
- Avoid using default router credentials
- Disable WPS if not needed
- Monitor Wi-Fi networks for suspicious activity



ADVANCED PERSISTENT THREATS (APT)

1.Advanced: Attackers use advanced malware and exploits and typically also have the skills and resources necessary to develop additional cyberattack tools and techniques, and may have access to sophisticated electronic surveillance equipment, satellite imagery, and even human intelligence assets.



ADVANCED PERSISTENT THREATS (APT)

2.Persistent: An APT may take place over a period of several years. The attackers pursue specific objectives and use a “low-and-slow” approach to avoid detection. The attackers are well organized and typically have access to substantial financial backing, such as a nation-state or organized criminal organization, to fund their activities.

3.**Threat:** An APT is deliberate and focused, rather than opportunistic. APTs are designed to cause real damage. including significant financial loss, destruction of systems and infrastructure, and physical harm and loss of life.